

Skills Summary

Second and Higher Derivatives

Use this reference while completing your worksheet or when studying.

1. Climbing the ladder: repeated differentiation

Definition: $f'' = (f')'$, $f''' = (f'')'$, $f^{(n)} = (f^{(n-1)})'$ — each rung is the derivative of the one above.

Polynomials: each rung lowers the degree by one, so a degree- n polynomial has $f^{(n)}$ constant and $f^{(n+1)} = 0$.

Simplify between rungs. An unsimplified derivative costs you twice on the next one.

Example: Find $f''(x)$ for $f(x) = x^4 - 5x^2$.

Step 1. Only the power rule is needed, so take one rung at a time and simplify before differentiating again.

Step 2. $f'(x) = 4x^3 - 10x$, then differentiate that.

Step 3. $f''(x) = 12x^2 - 10$

Try it: Find $g'''(x)$ for $g(x) = x^5 + 2x^3$.

check: $60x^2 + 12$

2. Second derivatives with the product and chain rules

The rules do not stop at the first rung. Whatever rule built f' , f' is a new function and needs its own rule.

Chain rule twice: for $f = \sin(kx)$, each differentiation contributes a factor of k , so $f'' = -k^2 \sin(kx)$.

Product rule: factor the common exponential or trig piece out of f' first — it turns two product-rule applications into one.

Example: Find $f''(x)$ for $f(x) = xe^x$.

Step 1. f is a product, so f' needs the product rule — and f' is another product, so the second rung needs it again.

Step 2. $f'(x) = e^x + xe^x = e^x(x + 1)$, so $f''(x) = e^x(x + 1) + e^x(1) = e^x(x + 2)$.

Step 3. $f''(x) = (x + 2)e^x$

Try it: Find $f''(x)$ for $f(x) = \cos(4x)$.

check: $-16\cos(4x)$

3. Implicit second derivatives

Two rungs, same rule. Differentiate the equation implicitly to get $\frac{dy}{dx}$, then differentiate *that* — and every y inside it is still a function of x , so it carries another $\frac{dy}{dx}$.

Finish by substituting: replace $\frac{dy}{dx}$ with the expression from the first rung, then use the original equation to simplify.

Example: For $x^2 + y^2 = 9$ on the upper arc, find $\frac{d^2y}{dx^2}$ in terms of x .

Step 1. The equation is not solved for y , so differentiate implicitly, then differentiate the result and substitute back.

Step 2. $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$; differentiating again gives $-\frac{y^2+x^2}{y^3} = -\frac{9}{y^3}$.

Step 3. On the upper arc $y = \sqrt{9 - x^2}$, so $\frac{d^2y}{dx^2} = \frac{-9}{(9 - x^2)^{3/2}}$

Try it: For $x^2 + y^2 = 16$ on the upper arc, find $\frac{d^2y}{dx^2}$ in terms of x .

check: $\frac{2x(2x - 9)}{9}$

4. What the second derivative means

Motion: $s(t)$ position, $s'(t) = v(t)$ velocity, $s''(t) = a(t)$ acceleration.

Shape: $f'' > 0$ concave up, $f'' < 0$ concave down. A sign change in f'' is an inflection point.

Setting $s''(t) = 0$ finds the instants where acceleration vanishes — the candidates for an inflection point on the position graph.

Example: For $s(t) = t^3 - 3t^2$ metres, find when the acceleration is zero.

Step 1. Acceleration is the second rung, so differentiate twice and set that equal to zero.

Step 2. $s'(t) = 3t^2 - 6t$, so $a(t) = s''(t) = 6t - 6$; solving $6t - 6 = 0$ gives the time.

Step 3. $t = 1$ second, where the position graph changes concavity.

Try it: For $s(t) = 2t^3 - 12t^2 + 5t$ metres, find when the acceleration is zero.

check: $t = 2$